

Leveraging Gamification to Address Child Sexual Abuse: A Preliminary Evaluation of the Cesagram Platform

Eftychia Roumelioti
Fondazione Bruno Kessler
Trento, Italy
eroumelioti@fbk.eu

Giulia Deppieri
Fondazione Bruno Kessler
Trento, Italy
gdeppieri@fbk.eu

Gianluca Schiavo
i3 unit
Fondazione Bruno Kessler
Trento, Italy
gschiavo@fbk.eu

Annapaola Marconi
Fondazione Bruno Kessler
Trento, Italy
marconi@fbk.eu

Abstract

Child Sexual Abuse (CSA) remains a critical global issue, with the rise of online grooming posing new threats to young people's safety. This paper presents the Cesagram platform, a gamified digital platform designed to prevent CSA by raising awareness of online grooming among children aged 11-14. The platform integrates interactive learning activities within a narrative-driven environment, featuring customizable avatars and four thematic districts. The effectiveness of the Cesagram platform combined with theoretical content was evaluated through 11 workshops involving 195 students from Lithuania and Greece, utilizing pre- and post-workshop questionnaires. Findings indicate that the workshops effectively improved participants' knowledge of CSA and grooming, while also promoting user engagement. This study underscores the potential of gamification in sensitive educational contexts, contributing to digital well-being and safety education.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Applied computing** → **Interactive learning environments**.

Keywords

Gamification, Child sexual abuse, Internet safety, Playful education, Sensitive topics

ACM Reference Format:

Eftychia Roumelioti, Gianluca Schiavo, Giulia Deppieri, and Annapaola Marconi. 2025. Leveraging Gamification to Address Child Sexual Abuse: A Preliminary Evaluation of the Cesagram Platform. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3706599.3706684>

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI EA '25, April 26–May 01, 2025, Yokohama, Japan

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3706684>

1 Introduction

Child Sexual Abuse (CSA) encompasses any form of sexual activity with a minor where consent is not legally valid, including direct acts such as physical abuse and indirect actions like exposure to explicit material, voyeurism, and grooming [8, 22]. Grooming involves a calculated process in which offenders build trust with a child to exploit them sexually, often through flattery, deception, or online communication [8]. CSA is alarmingly widespread, with 16.8% of women and 7.9% of men reporting such abuse before age 18, and a global prevalence rate estimated at 12%. These statistics highlight the urgent need for effective prevention strategies to address this pervasive issue [28].

The rise of digital technology has profoundly changed how perpetrators target potential victims, giving rise to new forms of abuse, such as cyber-grooming [3]. This method involves offenders using online platforms to manipulate and desensitize minors, often gradually introducing sexual content and interactions to establish control [8, 16]. The anonymity and accessibility of digital communication have expanded the reach and severity of these abuses, making detection and prevention increasingly challenging [35]. Consequently, there is a critical need for prevention strategies that effectively address both traditional and online grooming tactics.

To combat CSA and grooming, various initiatives have been developed to educate children and adolescents about the potential dangers in online environments [16, 21]. These initiatives typically include lessons on protecting personal information and strategies for avoiding unsolicited online interactions with strangers. Additionally, a key aspect involves equipping parents and caregivers with the tools and knowledge needed to recognize signs of grooming and CSA, empowering them to take preventive measures [7, 27].

One promising approach to enhance these educational efforts is gamification. The incorporation of game-based elements into educational content is well-known to enhance engagement, motivation, and knowledge retention [15]. Although gamification has demonstrated potential in various educational contexts, its application to CSA prevention is still relatively new and underexplored compared to other application domains [2, 21].

The Cesagram platform seeks to fill this gap by providing a gamified educational tool that complements educational approaches, engaging young people, educators, and caregivers in meaningful

learning experiences about CSA prevention. By incorporating insights from psychology, education, and gamification theory, the platform aims to create an inclusive, safe and empathetic environment that fosters understanding and proactive behaviours. This paper presents the design and development of the Cesagram platform, alongside an evaluation of its effectiveness when combined with theoretical content. The evaluation, conducted through 11 workshops with students from Lithuania and Greece, provided preliminary insights into how well the workshops enhanced participants' knowledge and awareness of CSA prevention.

2 Related Work

Research on technology-based interventions for preventing CSA has increasingly focused on a diverse range of digital tools, including serious games and virtual reality applications [21, 36]. These tools are designed to support both prevention and treatment efforts. Asadzadeh et al. provided a comprehensive review of digital interventions addressing various aspects of child abuse, highlighting the effectiveness of serious games in prevention by engaging children through interactive and enjoyable learning experiences [2]. Numerous studies have emphasized the importance of designing engaging content tailored to different audiences, including programs that promote gender equality and CSA prevention in educational settings like schools [23, 25, 37].

In therapeutic contexts, digital tools play a key role in supporting CSA survivors. For example, serious games such as "Vil Du?!" enable children to communicate their experiences non-verbally, easing the therapeutic process and making it less intimidating [10]. Storytelling applications further provide creative avenues for children to narrate their experiences in safe, supportive environments, helping to reduce anxiety and emotional distress while fostering empowerment and recovery [19].

Serious games have also demonstrated significant potential in CSA prevention through interactive scenarios that engage both children and adults. One prominent example is the "Orbit" game, which promotes relationship-building, trust, and self-esteem enhancement while raising awareness about CSA and encouraging children to disclose concerns to trusted adults [13, 25, 27]. Another notable example, "Hidden in the Park", uses decision-making scenarios to teach children about online risks, encouraging critical thinking and fostering digital safety awareness [29].

In addition to serious games, mobile applications have been developed to educate children and parents about CSA prevention. However, many of these CSA-related apps are adaptations of serious games that often fall short of meeting effective educational standards. Although some of these apps offer valuable learning experiences, few are designed with the comprehensiveness required for school-based prevention programs. Research indicates that improvements in both the quality of educational content and engagement strategies are needed to enhance the effectiveness of these mobile applications [20].

Despite the promise of serious games and mobile apps, the use of gamification in CSA prevention remains limited. Gamification involves the integration of game-like elements into non-game educational contexts and offers distinct advantages that can enhance educational efforts [6, 15, 31]. One key strength of gamification is its

adaptability; educators can seamlessly incorporate game elements into existing educational materials without the extensive resources typically required for developing serious games [1]. Additionally, gamification can be tailored to fit various educational contexts, facilitating personalized learning experiences that cater to diverse learning styles and scalable needs [18]. By incorporating game mechanics such as points, levels, badges, and challenges, gamification enhances engagement while providing students with a structured and approachable way to engage with sensitive topics [4, 24]. This is particularly valuable in areas like CSA prevention, where the nature of the topic can be emotionally challenging or intimidating.

While serious games and mobile applications have dominated technology-mediated interventions, the current literature on gamification in sensitive contexts like CSA prevention is still sparse, indicating a need for further exploration and research to unlock its potential benefits. Gamification's ability to offer flexible, scalable, and engaging educational experiences makes it a promising tool for CSA prevention efforts. This paper aims to address this gap by exploring the potential of gamification in complementing CSA education, assessing the effectiveness of this approach, and identifying key areas for future research.

3 The Cesagram Platform

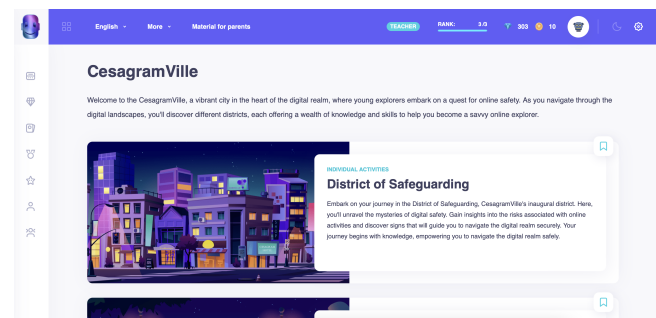


Figure 1: The Cesagram platform

The Cesagram Platform¹ is a gamified educational tool developed to tackle online safety and grooming, with a primary focus on students aged 11-14 and their caregivers (see Fig. 1). Developed as part of a European-funded project, the platform is available in four languages: English, Lithuanian, Greek, and Italian. It is specifically designed to be combined with traditional theoretical educational approaches, enhancing the learning process through interactive gamified elements. The platform's design was informed by a comprehensive investigation, including a literature review and consultations with professionals, to understand how gamification can complement education on CSA prevention, as discussed in [24]. The educational content used in the platform is based on a training curriculum developed by the non-profit organisation CESIE for CSA education² and anonymized real-life chat logs from online grooming cases provided by the project's partners.

¹<https://cesagramplatform.eu/>

²<https://cesie.org/en/resources/cesagram-training-curriculum/>

The following sections outline the platform's core features, illustrating how these insights and content have been applied to create an engaging and educational experience for the users.

3.1 Educational Activities

The platform is structured into four districts within "CesagramVille" that users have to explore, each designed to address different aspects of online safety and present to users specific educational activities:

District of Safeguarding: This district is dedicated to helping users recognize and understand signs of CSA and grooming. The activities are crafted to enhance awareness of online threats and their potential links to children going missing. For instance, users might engage in tasks that involve matching terms related to CSA with their definitions or exploring interactive stories that illustrate grooming tactics and their implications.

Self-Awareness Park: In this district, users delve into concepts of personal boundaries, consent, and emotional intelligence. Activities are designed to foster reflection on distinguishing healthy from unhealthy relationships and to develop strategies for maintaining personal safety. Examples include exercises that involve categorizing actions as either safe or unsafe and fill-in-the-blank tasks that teach the principles of digital consent.

Prevent & Respond Alley: This district focuses on equipping users with practical strategies for preventing and responding to grooming situations. The activities aim to build skills for recognizing warning signs and making informed decisions in potentially dangerous scenarios. Notable examples include identifying "red flags" in a simulated online conversation between a child and a stranger (see Fig. 2) and selecting the safest response from a list of messages in a chat simulation.

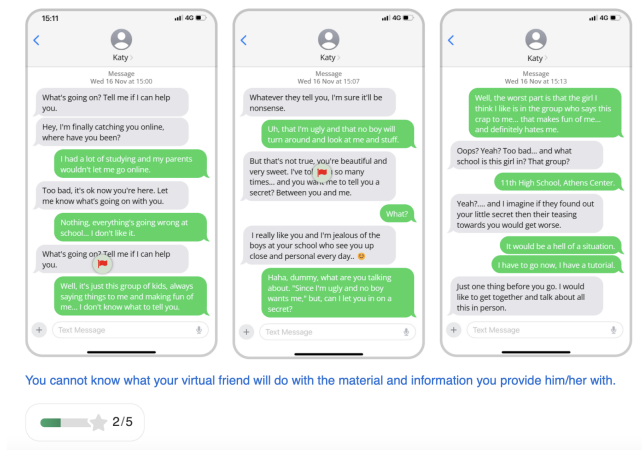


Figure 2: Activity in the "Prevent & Respond Alley" of the Cesagram platform

Unity Square: The final district provides opportunities for group activities and collaborative learning under the supervision of an educator. Accessible only to educators, Unity Square features activities that encourage students to work together, reflect on collective strategies for addressing grooming, and discuss their insights with

peers. These group tasks aim to foster a sense of community and enhance learning through collaboration.

In addition to these district-based activities, the platform includes resources and activities specifically designed for parents and caregivers. These resources are embedded within the platform to help parents actively support their children's learning about online safety.

3.2 Gamification Design

In the gamification design of the Cesagram platform, the Octalysis framework was used as a reference in selecting game elements to enhance user engagement [5, 30]. Given the framework's wide range of elements, the list was refined based on specific criteria. A key objective was to avoid competitive elements or features that might provoke negative emotions [26], in line with Chou's concept of Black Hat cores. Instead, the platform focuses on positive reinforcement and collaborative learning, as in [11].

Narration plays a central role in engaging users and guiding their experience. The platform is built around the virtual world of **CesagramVille** which features various districts each representing a key aspect of online safety and CSA awareness. The virtual city encourages users to explore and interact with the districts, making the learning experience both enjoyable and educational. As they progress through activities, users earn two types of points: **Group Wisdom** and **Cyber Credits**. Group Wisdom points are accumulated collectively by all participants in a class, including parents who contribute by completing specific activities for adults. Cyber Credits, earned on an individual basis, can be used to customize avatars. This dual-point system encourages personal achievements while fostering a sense of collective progress, promoting both individual and group engagement.

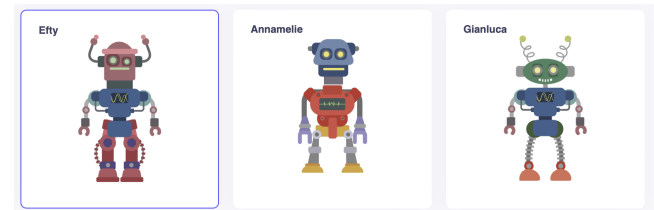


Figure 3: Customized avatars by a group of users

Avatar customization is a central feature of the platform. As users accumulate Group Wisdom points collectively, they unlock new equipment options for their avatars. They can then use their Cyber Credits to acquire this equipment, which provides a tangible sense of accomplishment and progression. This feature not only enhances individual engagement but also fosters a sense of community as users view and compare avatars within their group, driving friendly competition and motivation (see Fig. 3).

The platform is also structured around **missions** and **ranks** (see Fig. 4). Missions such as "Secure Navigation" and "Emotional Intelligence" challenge users to complete activities in each district. Users progress through different ranks—Beginner, Explorer, and Mastermind—by collecting more Group Wisdom points. Advancing through these ranks not only celebrates users' achievements but

also rewards them with additional Cyber Credits, incentivizing further participation.

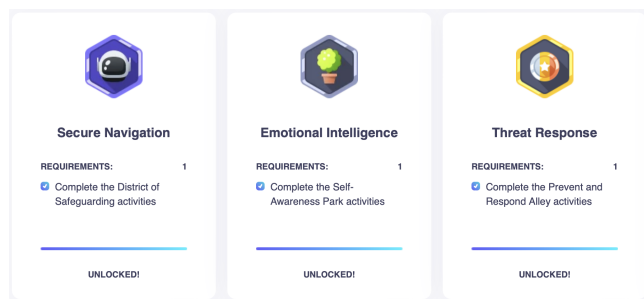


Figure 4: Examples of missions of the Cesagram platform

4 Workshops

A total of 11 workshops were conducted between April and June 2024, including 9 workshops in Lithuania (Šalčininkai, Širvintos, and Vilnius) and 2 workshops in Athens, Greece (see Fig.5). Further details are provided below.

4.1 Research Questions

The case study aimed to investigate whether the Cesagram platform, when combined with theoretical content, effectively raises awareness of CSA among students. Additionally, the study sought to gather insights into students' perceptions of the platform. These objectives were translated into the following research questions:

RQ1. How are workshops supported by the Cesagram platform effective in raising students' awareness of CSA?

RQ2. What are students' perceptions of the Cesagram platform?

4.2 Participants & Setting

A total of 195 students, aged 10 to 18 years ($M = 12.49$, $SD = 1.74$), were recruited from schools and youth centres in different cities in Lithuania (Šalčininkai, Širvintos, Vilnius) and in Greece (Athens). The recruitment process was facilitated by municipalities, which shared information about the pilot training with relevant institutions. Among the students, 91 identified as female, 87 as male, 7 as another gender, and 10 chose not to disclose their gender. A total of 9 workshops were conducted in various cities across Lithuania, with each workshop group comprising approximately 15 students, and 2 workshops were conducted in Athens, each composed of approximately 22 students.

All participants took part on a voluntary basis, with parental consent obtained through written forms approved by the ethical committee of CESIE. The consent process was managed in collaboration with teachers, youth workers, and psychologists, who also coordinated logistical aspects, including declarations.

4.3 Data Collection

Data collection during the workshops was conducted in adherence to CESIE's Child Protection Policy³, ensuring full anonymity

³<https://cesie.org/en/child-protection-policy/>



Figure 5: Workshop with the Cesagram platform in Athens

for all participants. Participants' awareness of CSA was measured using pre- and post-workshop questionnaires. Additionally, after the workshops, students were asked to provide feedback on their experiences with the gamified platform through post-workshop questionnaires. All collected data was anonymized and securely stored to ensure participant confidentiality.

Given the exploratory nature of the study, the specific context of the case study, and the diverse range of targeted young individuals, a tailored questionnaire was developed to address RQ1. The questionnaire aimed to measure key concepts covered in CSA prevention education programs and included constructs designed to assess knowledge and attitude gains post-program. Drawing inspiration from established instruments such as the Children's Knowledge of Abuse Questionnaire (CKAQ) [33], the questionnaire comprised 9 items targeting common misconceptions regarding CSA. Examples of these items include statements like "Sometimes it is possible to do things that you are not completely sure about, just to make adults happy" and "Violence is limited to physical harm". Additionally, questions related to online grooming and technology-mediated CSA were also incorporated, e.g. "It is not dangerous to meet in person people you've met online" and "Social media and messaging platforms are completely safe spaces". This questionnaire allowed for the assessment of changes in students' knowledge and attitudes, providing insights into the platform's educational effectiveness.

To address RQ2, questions were selected based on the Technology Acceptance Model (TAM) [9], covering antecedents and outcomes such as perceived ease of use, perceived usefulness, intention to use, and social influences. Additionally, the assessment included questions related to satisfaction, perceived engagement and fun with the platform, and the overall impact on knowledge and behavior regarding child sexual abuse (CSA) topics.

4.4 Protocol

The workshops followed a structured protocol, detailed in Table 1. The protocol was designed to ensure consistency in the delivery of activities across different groups and locations while maintaining flexibility to address the specific needs of participants.

Duration	Activity
15 min	Icebreaker activity
15 min	Pre-evaluation questionnaire
40 min	Module 1
	Theory on identifying grooming and missing related dangers
	Activities on the platform
40 min	Module 2
	Theory on raising awareness on comprehensive sex education and safer online space
	Activities on the platform
40 min	Module 3
	Theory on preventing and responding to online grooming
	Activities on the platform
15 min	Closing session
15 min	Post-evaluation questionnaire

Table 1: Protocol of the Cesagram workshops

5 Results

The majority of participants (98%) reported having played a digital game with an educational purpose at least once. When asked about the frequency of playing educational games, 32% reported playing very rarely, 9% once a year, 36% every six months, 17% monthly, and 23% weekly. 68% of participants reported being familiar with the term “grooming”.

To ensure complete anonymity, questionnaire data were analyzed as independent groups rather than pairwise. The results were thus examined using a t-test for independent samples. The scale on knowledge regarding CSA false myths reached a reliability of $\alpha = 0.73$ and the scores were averaged across all nine items. The analysis shows that scores were significantly higher in post-training ($M=2.6$, $SD=0.79$) compared to pre-training ($M=2.1$, $SD=0.73$), with an average increase of 0.5 on a five-point scale ($t(388)=-5.5$, $p < .001$).

To assess potential gender effects, an ANOVA was conducted with two between-group factors: “pre-post condition” and “gender”. The results revealed a significant interaction between the factors ($F(1,345)=15.5$, $p < .001$) and a significant main effect of the pre-post condition ($F(1,345)=31.7$, $p < .001$). Post-hoc analysis indicated that the pre-post difference was significant for female participants ($p < .001$), while the effect for male participants was not statistically significant ($p = 0.638$). Additionally, post-training scores were higher for females than males ($p < .001$).

Age differences were also analyzed by comparing results across three age groups: participants aged 10–12 years (56%), 13–14 years (34%), and over 14 years (10%). An ANOVA with “pre-post condition” and “age group” factors showed neither a significant interaction ($p = 0.076$) nor a significant effect of age group ($p = 0.053$).

Scores on the platform’s experience was generally positive, with all scores significantly different from the midpoint of the scale (middle point equal to 3; except for the item “I would use this platform outside of school in my own free time” - see Table 2). No

significant differences in platform experience were observed based on gender ($p = 0.496$) or age group ($p = 0.528$).

6 Discussion

As far as RQ1 on the platform’s effectiveness is concerned, the results of the case study demonstrate that Cesagram workshops can effectively enhance knowledge about false myths surrounding CSA. Participants showed a significant increase in awareness after engaging with the platform, underscoring its potential as a valuable educational tool. The observed positive impact on knowledge gains aligns with prior studies that highlight the importance of interactive and gamified learning environments in fostering a deeper understanding of complex and sensitive topics like CSA [14].

A notable finding from the gender analysis indicates that female participants benefitted more from the training compared to males. While the platform successfully engages many users, this gender disparity suggests the need for tailored content or delivery methods to better resonate with male users. For instance, incorporating CSA-related scenarios that reflect male experiences or concerns more directly could increase relevance [34]. Additionally, gamification elements that are particularly motivating for male users, such as progress-based rewards or team-oriented challenges, could enhance engagement [32]. On the other hand, the absence of significant differences across age groups implies that the platform’s content is versatile enough to cater to a diverse range of developmental stages beyond its targeted age range of 11–14 years.

Regarding RQ2 on students’ perceptions, the results reveal a generally positive response to the platform’s educational value and ability to engage participants. This is evidenced by significantly positive scores across most experience-related items, indicating that students found the platform engaging and effective for learning. However, the lower intention to use the platform outside of school suggests a need for strategies to encourage continued engagement in informal settings. Potential enhancements could include integrating features such as personalized challenges, social interaction elements, or reward systems that promote long-term use and independent exploration [12].

Although the findings are promising, certain limitations must be acknowledged. The lack of a control group limits the ability to draw definitive conclusions about the platform’s standalone impact. To improve the reliability and generalizability of the findings, future research should include a more diverse participant pool and incorporate controlled experiments, comparing workshops that use the platform with those that rely solely on theoretical instruction. Additionally, evaluating a non-gamified version of the platform could provide insights into the specific contribution of gamification to learning outcomes.

Several directions for future work emerge from this study. Enhancing customization options, such as personalized learning paths and adaptive difficulty levels, could better address diverse user needs. Expanding the platform’s scope to cover other sensitive topics, such as gender-based violence or online safety, would broaden its applicability. Cultural adaptations of the platform, co-designed with local educators and experts, are essential to ensure the platform remains relevant across different educational and sociocultural contexts. Furthermore, future iterations could incorporate more

Dimension	Question	Score (M, SD)
Fun	The platform was both fun and educational	M = 4.10, SD = 1.11
Learning	I have learned new things from using the platform	M = 3.84, SD = 1.19
Usefulness	I think that this platform is a useful learning tool for people my age	M = 3.93, SD = 1.12
Awareness	This platform has made me reconsider things I've said or done	M = 3.47, SD = 1.24
Behavioural change	I will try to change my behavior or perceptions as a result of using this platform	M = 3.47, SD = 1.25
Context of usage	I would use this platform outside of school in my own free time	M = 3.32, SD = 1.25
Future usage	I would use this platform again in the future	M = 3.39, SD = 1.35
Recommendation	I would recommend this platform to my friends	M = 3.78, SD = 1.12
Average	[Averaged score - $\alpha = 0.87$]	M = 3.66, SD = 0.89

Table 2: Feedback from Young Participants on the Platform

immersive elements, such as virtual reality experiences or collaborative challenges.

7 Conclusions

This paper presented the design and development of the Cesagram platform, a gamified educational tool aimed at raising awareness of CSA among children aged 11-14 and their parents. The platform integrates critical educational content on CSA prevention with gamification techniques such as points, levels, and badges, creating an engaging and interactive learning environment. By combining these elements, the Cesagram platform seeks to foster deeper understanding and retention of sensitive topics, offering a novel way to engage young learners and their families in important discussions around CSA.

A preliminary evaluation of the Cesagram workshops was conducted through sessions with 195 students in Lithuania and Greece, where the platform was used alongside theoretical teachings on CSA. The results of the evaluation revealed promising findings, demonstrating that the workshops were effective in raising awareness about CSA and dispelling myths surrounding it. The positive reception from participants suggests that the platform successfully engages students, motivates them to learn, and enhances their knowledge of CSA prevention.

Overall, the positive findings from this case study, even though preliminary, pave the way for using gamification to address CSA and other sensitive topics [11, 17]. These findings show that engaging educational tools have the potential to improve awareness and understanding among different groups, highlighting gamification's potential to encourage meaningful discussions about sensitive and often stigmatised issues. Future work can build on these insights by refining the platform, expanding its scope to cover additional sensitive topics, and exploring its adaptability across different cultural and educational contexts.

Acknowledgments

The Cesagram platform is part of the European Union-funded project Cesagram with project number 101084974. We would like to express our gratitude to CESIE for their work in developing the training curriculum on which the Cesagram platform activities were based. We also extend our thanks to "Dingusių žmonių šeimų

paramos centras" and "The Smile of the Child" for organizing and facilitating the workshops in Lithuania and Greece, respectively.

References

- [1] Rula Al Azawi, Mazin Bulshi, and Fatma farsi. 2016. Educational Gamification Vs. Game Based Learning: Comparative Study. *International Journal of Innovation, Management and Technology (IJIMT)* 7 (08 2016), 131–136. <https://doi.org/10.18178/ijimt.2016.7.4.659>
- [2] Afsoon Asadzadeh, Hassan Shahrokhi, Behzad Shalchi, Zhila Khamnian, and Peyman Rezaei-Hachesu. 2022. Digital games and virtual reality applications in child abuse: A scoping review and conceptual framework. *PLOS ONE* 17, 11 (Nov. 2022), e0276985. <https://doi.org/10.1371/journal.pone.0276985>
- [3] W. Binford. 2023. What Frontline Professionals Need to Combat Child Maltreatment Online. *International Journal on Child Maltreatment: Research, Policy and Practice* 6, 2 (2023), 333–339. <https://doi.org/10.1007/s42448-023-00164-x>
- [4] KOO BonJin and WOO HoSung. 2024. Using Gamification Development of Sex Education Program (Ethical Perspective) for Youth. *Journal of Research and Publication Ethics* 5, 1 (2024), 19–27.
- [5] Yu-kai Chou. 2015. The octalysis framework for gamification & behavioral design. URL: [https://yukaichou.com/gamification-examples/octalysis-complete-gamification-framework/\[accessed 2019-11-03\]](https://yukaichou.com/gamification-examples/octalysis-complete-gamification-framework/[accessed 2019-11-03]) (2015).
- [6] Yu-kai Chou. 2019. *Actionable Gamification: Beyond Points, Badges, and Leaderboards*. Packt Publishing Ltd. Google-Books-ID: 9ZfBDwAAQBAJ.
- [7] Melissa Cowart-Osborne, Matthew Jackson, Elizabeth Chege, Evander Baker, Daniel Whitaker, and Shannon Self-Brown. 2014. Technology-Based Innovations in Child Maltreatment Prevention Programs: Examples from SafeCare®. *Social Sciences* 3, 3 (Aug. 2014), 427–440. <https://doi.org/10.3390/socsci3030427>
- [8] Samantha Craven, Sarah Brown, and Elizabeth Gilchrist. 2006. Sexual grooming of children: Review of literature and theoretical considerations. *Journal of Sexual Aggression* 12, 3 (Nov. 2006), 287–299. <https://doi.org/10.1080/13552600601069414>
- [9] F.D. Davis and A. Granić. 2024. *The Technology Acceptance Model: 30 Years of TAM*. Springer International Publishing, Imprint: Springer. <https://books.google.gr/books?id=L2X8EAAQBAJ>
- [10] Joyce Johanna Endendijk, Henny Tichelaar, Menno Deen, and Maja Deković. 2021. A Therapeutic Game for Sexually Abused Children and Adolescents (Vil Du?!): Exploratory Mixed Methods Evaluation. *JMIR Serious Games* 9, 3 (Aug. 2021), e26062. <https://doi.org/10.2196/26062>
- [11] Federica Gini, Eftychia Roumelioti, Fabio Fiori, Antonia Laura Philipa Jakobi, Boglarka Nyúl, Annapaola Marconi, Maria Paola Paladino, Gianluca Schiavo, and Massimo Zancanaro. 2024. Tackling Gender-Based Violence Through Gamification: a Preliminary Evaluation. In *Proceedings of the 2024 International Conference on Advanced Visual Interfaces (Arenzano, Genoa, Italy) (AVI '24)*. Association for Computing Machinery, New York, NY, USA, Article 42, 9 pages. <https://doi.org/10.1145/3656650.3656696>
- [12] Yusra Jadallah, Najwa Ahmad Khasawneh, Mohamad Ahmad, Mohammad Khasawneh, and Yusra Khasawneh. 2024. International Journal of Data and Network Science Exploring the long-term effects: Retention and transfer of skills in gamified learning environment. 20 (01 2024), 195–200. <https://doi.org/10.5267/j.ijdns.2023.10.004>
- [13] Christian Jones. 2008. Online games-based child safety environment. In *Proceedings of the 5th Australasian Conference on Interactive Entertainment (IE '08)*. Association for Computing Machinery, New York, NY, USA, 1–6. <https://doi.org/10.1145/1514402.1514406>
- [14] Christian Jones, Laura Scholes, Ben Rolfe, and Colleen Stieler-Hunt. 2020. A serious-game for child sexual abuse prevention: An evaluation of Orbit. *Child*

- Abuse & Neglect* 107 (2020), 104569. <https://doi.org/10.1016/j.chiabu.2020.104569>
- [15] Dimitrios N. Karagiorgas and Shari Niemann. 2017. Gamification and Game-Based Learning. *Journal of Educational Technology Systems* 45, 4 (June 2017), 499–519. <https://doi.org/10.1177/0047239516665105> Publisher: SAGE Publications Inc.
 - [16] Miljana Mladenović, Vera Ošmjanski, and Staša Vujičić Stanković. 2022. Cyber-aggression, Cyberbullying, and Cyber-grooming: A Survey and Research Challenges. *Comput. Surveys* 54, 1 (Jan. 2022), 1–42. <https://doi.org/10.1145/3424246>
 - [17] Boglárka Nyúl, Maria P Paladino, Antonia L P Jakobi, Federica Gini, Annapaola Marconi, Eftychia Roumelioti, Gianluca Schiavo, Jeroen Vaes, and Massimo Zancanaro. 2024. A Multidisciplinary Framework for Developing a Gamified Digital Platform to Combat Gender-based Violence. In *Gender R-Evolutions: Imagining the Inevitable, Subverting the Impossible*. Centre for Interdisciplinary Gender Studies, University of Trento.
 - [18] Manuel Oliveira and Sobah Petersen. 2014. The Choice of Serious Games and Gamification. In *Serious Games Development and Applications*, Minhua Ma, Manuel Fradinho Oliveira, and Jannicke Baalsrud Hauge (Eds.). Springer International Publishing, Cham, 213–223.
 - [19] Amita Pharshy. 2016. Children's Storytelling App For Detecting Potential Child Sexual Abuse. <https://openresearch.ocadu.ca/id/eprint/734/> Num Pages: 78 Publisher: OCAD University.
 - [20] Sadia Tasnuva Pritha, Rahnuma Tasnim, Muhammad Ashad Kabir, Sumaiya Amin, and Anik Das. 2023. A Systematic Review of Mobile Apps for Child Sexual Abuse Education: Limitations and Design Guidelines. *International Journal of Mobile Learning and Organisation* 1, 1 (2023), 1. <https://doi.org/10.1504/IJMLO.2023.10049097> arXiv:2107.01596 [cs].
 - [21] Sadia Tasnuva Pritha, Rahnuma Tasnim, Muhammad Ashad Kabir, Sumaiya Amin, and Anik Das. 2024. Smartphone apps for child sexual abuse education: gaps and design considerations. *Int. J. Mob. Learn. Organ.* 18, 1 (jan 2024), 101–134. <https://doi.org/10.1504/ijmlo.2024.135124>
 - [22] Frank W. Putnam. 2003. Ten-Year Research Update Review: Child Sexual Abuse. *Journal of the American Academy of Child & Adolescent Psychiatry* 42, 3 (March 2003), 269–278. <https://doi.org/10.1097/00004583-200303000-00006>
 - [23] Eftychia Roumelioti, Federica Gini, Antonia Laura Philipa Jakobi, Annapaola Marconi, Boglárka Nyúl, Maria Paola Paladino, Gianluca Schiavo, and Massimo Zancanaro. 2023. StandByMe: A Gamified Educational Platform to Raise Awareness on Gender-Based Violence. In *Companion Proceedings of the Annual Symposium on Computer-Human Interaction in Play* (Stratford, ON, Canada) (*CHI PLAY Companion '23*). Association for Computing Machinery, New York, NY, USA, 108–113. <https://doi.org/10.1145/3573382.3616084>
 - [24] Gianluca Schiavo, Eftychia Roumelioti, Giulia Deppieri, and Annapaola Marconi. 2024. Gamification Strategies for Child Protection: Best Practices for Applying Digital Gamification in Child Sexual Abuse Prevention. In *Proceedings of the 23rd Annual ACM Interaction Design and Children Conference* (Delft, Netherlands) (*IDC '24*). Association for Computing Machinery, New York, NY, USA, 282–289. <https://doi.org/10.1145/3628516.3655813>
 - [25] Laura Scholes, Christian Jones, Colleen Stieler-Hunt, and Ben Rolfe. 2014. Serious games for learning: games-based child sexual abuse prevention in schools. *International Journal of Inclusive Education* 18, 9 (Sept. 2014), 934–956. <https://doi.org/10.1080/13603116.2013.860195>
 - [26] Sepandar Sepehr and Milena Head. 2013. Competition as an element of gamification for learning: an exploratory longitudinal investigation. In *Proceedings of the First International Conference on Gameful Design, Research, and Applications* (Toronto, Ontario, Canada) (*Gamification '13*). Association for Computing Machinery, New York, NY, USA, 2–9. <https://doi.org/10.1145/2583008.2583009>
 - [27] Colleen Stieler-Hunt, Christian M. Jones, Ben Rolfe, and Kay Pozzebon. 2014. Examining key design decisions involved in developing a serious game for child sexual abuse prevention. *Frontiers in Psychology* 5 (2014). <https://doi.org/10.3389/fpsyg.2014.00073>
 - [28] Marije Stoltenborgh, Marinus H. Van IJzendoorn, Eveline M. Euser, and Marian J. Bakermans-Kranenburg. 2011. A Global Perspective on Child Sexual Abuse: Meta-Analysis of Prevalence Around the World. *Child Maltreatment* 16, 2 (May 2011), 79–101. <https://doi.org/10.1177/1077559511403920>
 - [29] Tarja Susi and Niklas Torstensson. 2019. “Who’s Texting?” – Playful Game Experiences for Learning to Cope with Online Risks. In *HCI in Games (Lecture Notes in Computer Science)*, Xiaowen Fang (Ed.). Springer International Publishing, Cham, 427–441. https://doi.org/10.1007/978-3-030-22602-2_32
 - [30] Ferran Teixes. 2017. Yu-Kai Chou (2016). Actionable Gamification: beyond points, badges and leaderboards. Octalysis Media: Fremont. CA. *RIO: Revista Internacional de Organizaciones* 18 (2017), 137–144.
 - [31] Armando M. Toda, Ana C. T. Klock, Wilk Oliveira, Paula T. Palomino, Luiz Rodrigues, Lei Shi, Ig Bittencourt, Isabela Gasparini, Seiji Isotani, and Alexandra I. Cristea. 2019. Analysing gamification elements in educational environments using an existing Gamification taxonomy. *Smart Learning Environments* 6, 1 (Dec. 2019), 16. <https://doi.org/10.1186/s40561-019-0106-1>
 - [32] Armando Maciel Toda, Wilk Oliveira, Lei Shi, Ig Ibert Bittencourt, Seiji Isotani, and Alexandra I. Cristea. 2019. Planning Gamification Strategies based on User Characteristics and DM: A Gender-based Case Study. *CoRR abs/1905.09146* (2019). arXiv:1905.09146 <http://arxiv.org/abs/1905.09146>
 - [33] Leslie M. Tutty. 2020. Children’s Knowledge of Abuse Questionnaires (CKAQ)-Short: Two Brief Ten-Item Measures of Knowledge about Child Sexual Abuse Concepts. *Journal of Child Sexual Abuse* 29, 5 (2020), 513–530. <https://doi.org/10.1080/10538712.2019.1688443>
 - [34] Mary M. Wellman. 1993. Child sexual abuse and gender differences: Attitudes and prevalence. *Child Abuse & Neglect* 17, 4 (1993), 539–547. [https://doi.org/10.1016/0145-2134\(93\)90028-4](https://doi.org/10.1016/0145-2134(93)90028-4)
 - [35] Rebecca Williams, Ian A. Elliott, and Anthony R. Beech. 2013. Identifying Sexual Grooming Themes Used by Internet Sex Offenders. *Deviant Behavior* 34, 2 (Feb. 2013), 135–152. <https://doi.org/10.1080/01639625.2012.707550>
 - [36] Putri Prevya Yanti, Yusep Rosmansyah, and Budiman Dabarsyah. 2019. Serious Games for Children: A Systematic Literature Review. In *2019 International Conference on Informatics, Multimedia, Cyber and Information System (ICIMCIS)*. 79–84. <https://doi.org/10.1109/ICIMCIS48181.2019.8985230>
 - [37] Alma Gloria Barrera Yañez, Cristina Alonso-Fernández, and Baltasar Fernández-Manjón. 2023. Systematic literature review of digital resources to educate on gender equality. *Education and Information Technologies* 28, 8 (Aug. 2023), 10639–10664. <https://doi.org/10.1007/s10639-022-11574-8>